AWS S3 Static Website Hosting

Week 2 – Cloud Computing Internship Assignment (Task 2)

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AWS service	Amazon S3
Region	Asia Pacific (Mumbai) ap-south-1
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1. Objective

Host a static website on Amazon S3: store the website files in a bucket, enable static website hosting, allow public read access to the website objects, and verify the site through the S3 website endpoint.

2. Summary of the Setup

Item	Value
Bucket name	manish-static-site-111026
Region	Asia Pacific (Mumbai) ap-south-1
Bucket created	October 11, 2026, 11:39:27 (UTC+05:30)
Hosting type	Bucket hosting
Index document	index.html
Error document	error.html
Website endpoint	http://manish-static-site-111026.s3-website.ap-south-1.amazonaws.com
Access method	Bucket policy allowing s3:GetObject

3. Website Files

Three files were uploaded on October 11, 2026 at 11:46:50 (UTC+05:30). The site uses only HTML and CSS, so it can be served as a static website with no server-side processing.

File	Type	Size	Purpose
index.html	HTML	2.1 KB	Main homepage
style.css	CSS	2.4 KB	Website styling
error.html	HTML	935 B	Page shown when a page is not found

4. Implementation Steps

Step 1 – Create the S3 bucket

In the Amazon S3 console, a general purpose bucket named **manish-static-site-111026** was created in the Mumbai region (ap-south-1). Bucket names are globally unique, so the name carries a date-style suffix.

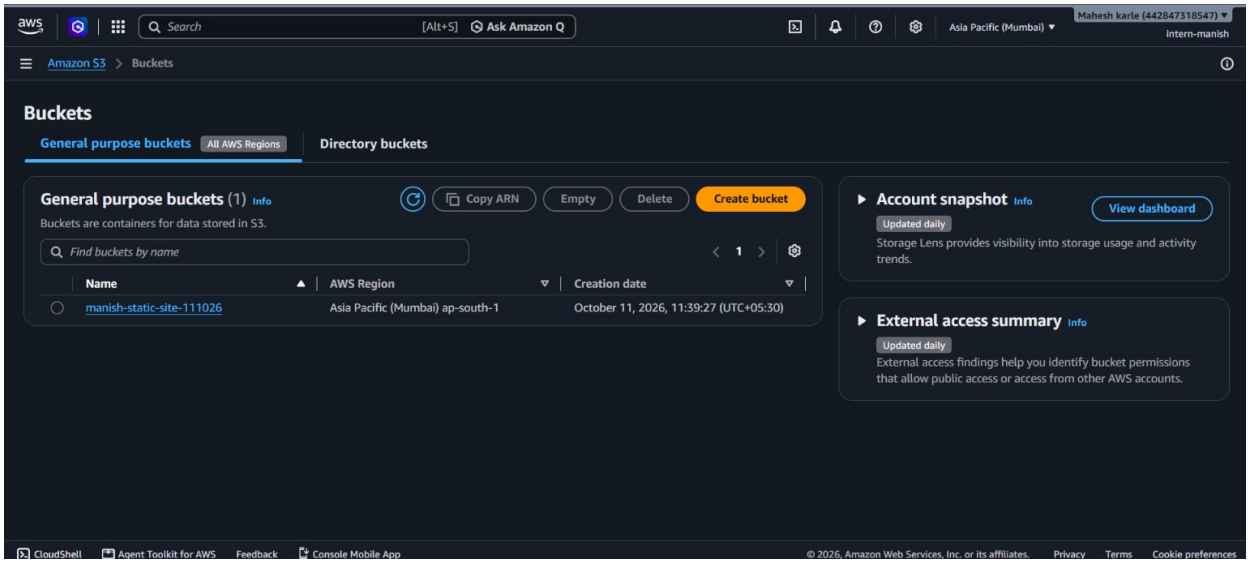


Figure 1: Bucket list showing manish-static-site-111026 in ap-south-1

Step 2 – Upload the website files

On the bucket's **Objects** tab, choose **Upload** and add index.html, style.css and error.html. The bucket then lists 3 objects in the Standard storage class.

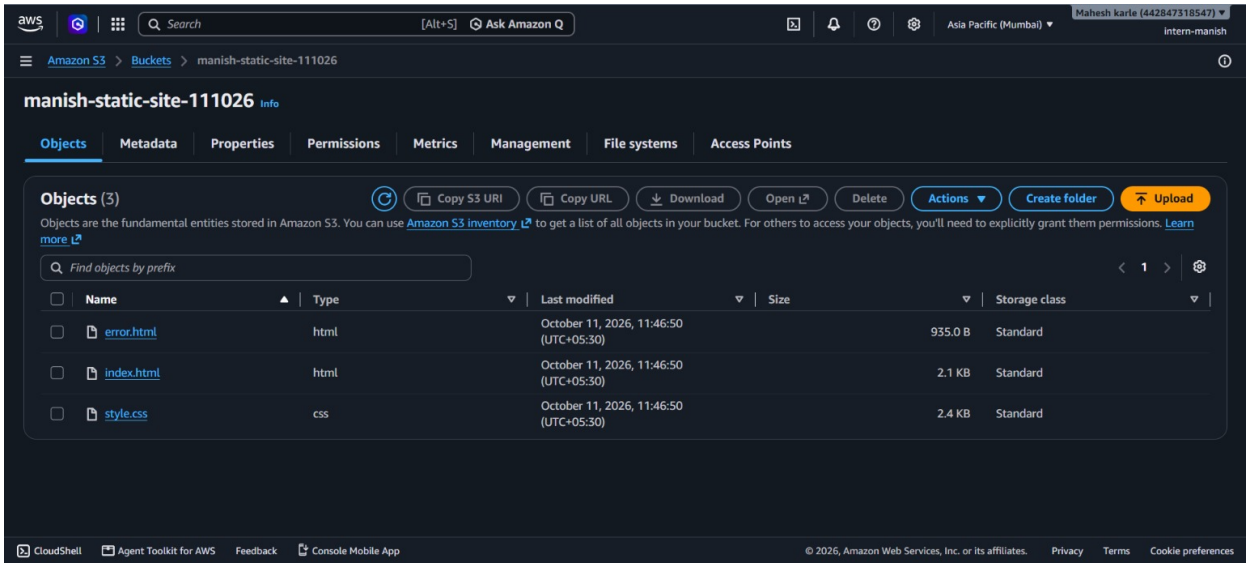


Figure 2: Uploaded objects – error.html, index.html and style.css

Step 3 – Enable static website hosting

Open the **Properties** tab, scroll to **Static website hosting** and choose **Edit**. Enable hosting, select **Host a static website**, then set the index document to **index.html** and the error document to **error.html**. After saving, the console shows hosting as Enabled, hosting type Bucket hosting, and the website endpoint. (The console's Amplify recommendation is informational only.)

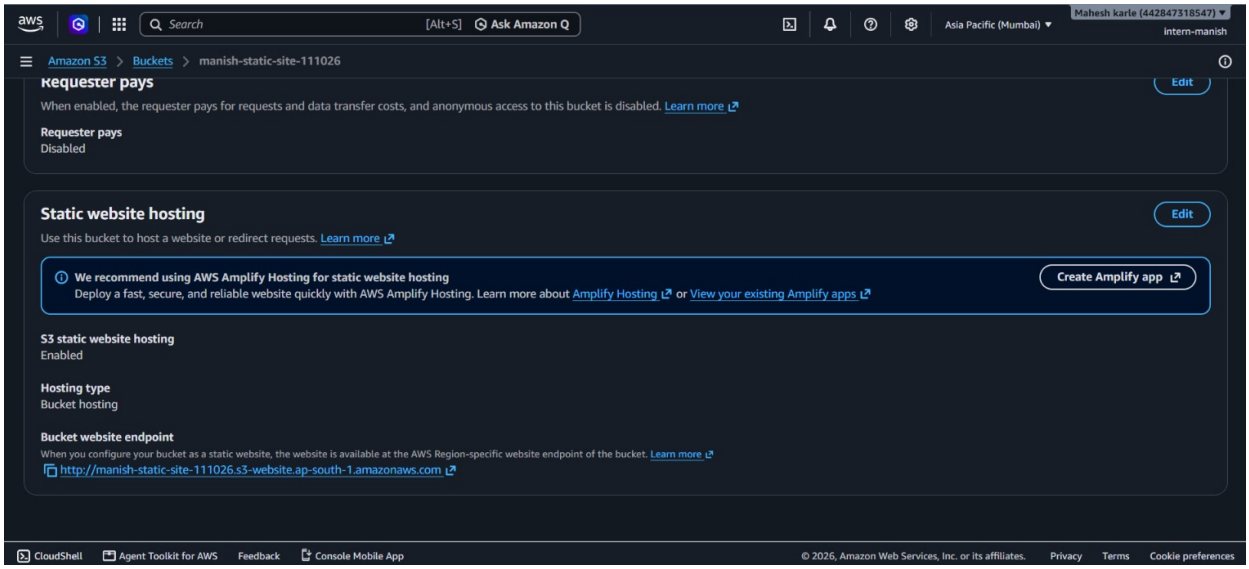


Figure 3: Static website hosting enabled with the bucket website endpoint

Step 4 – Allow public read access with a bucket policy

Visitors access the site anonymously, so the objects must be publicly readable. This requires the bucket's **Block public access** settings to permit public bucket policies. The following policy was added under **Permissions** → **Bucket policy**:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadForStaticWebsite",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::manish-static-site-111026/*"
    }
  ]
}
```

- **Effect Allow / Principal "*":** grants access to anyone.
- **Action s3:GetObject:** read-only; visitors cannot list, upload, modify or delete objects.
- **Resource .../*:** applies to every object in the bucket.

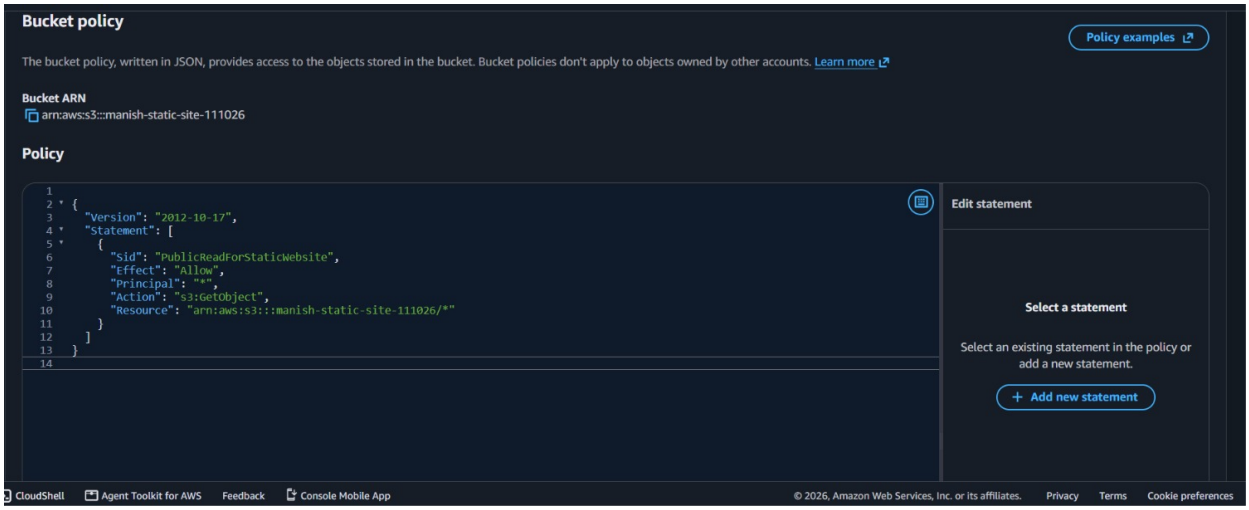


Figure 4: Public read bucket policy

Step 5 – Test the website

The website endpoint was opened in a browser and the site loaded correctly, with full CSS styling. This confirms that both the HTML and the style.css files are publicly readable.

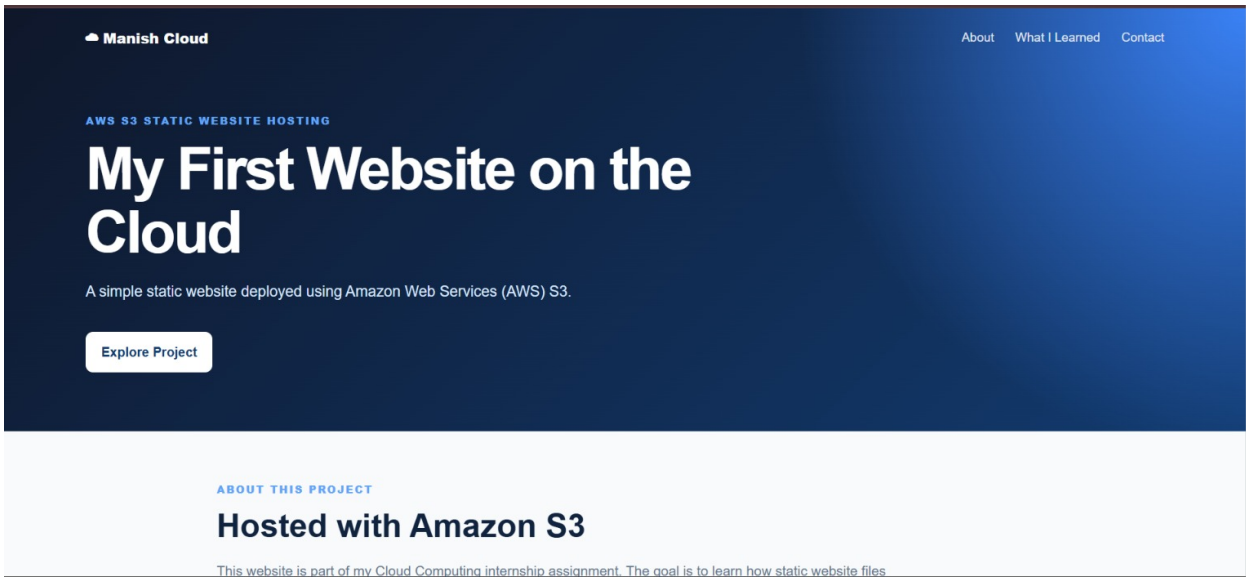


Figure 5: Live website – hero section, "My First Website on the Cloud"

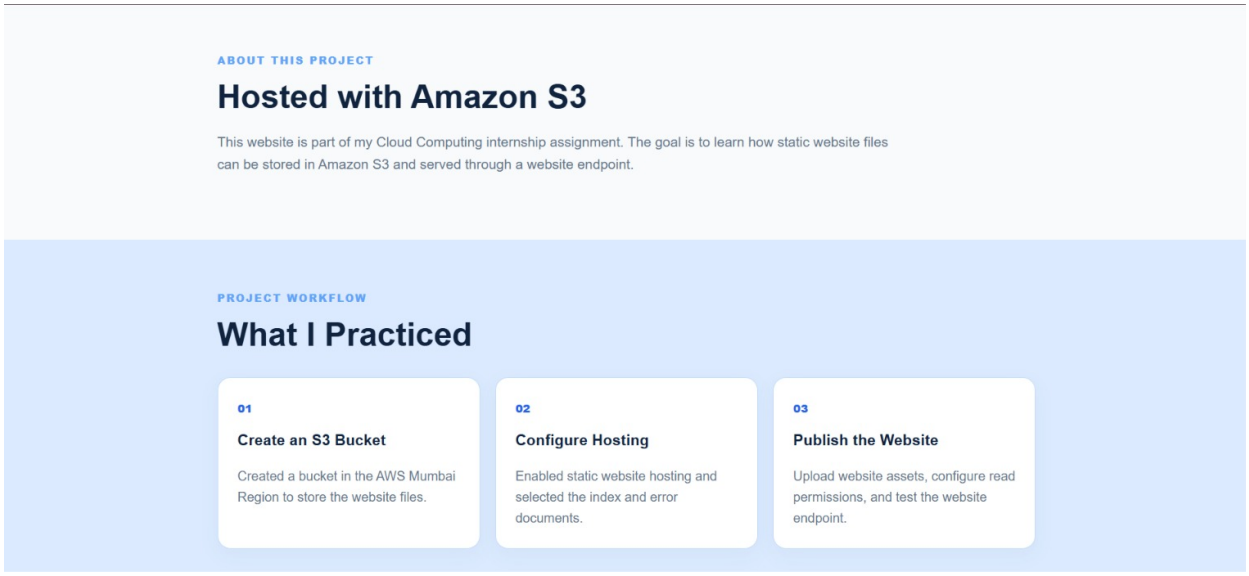


Figure 6: Live website – About and "What I Practiced" sections

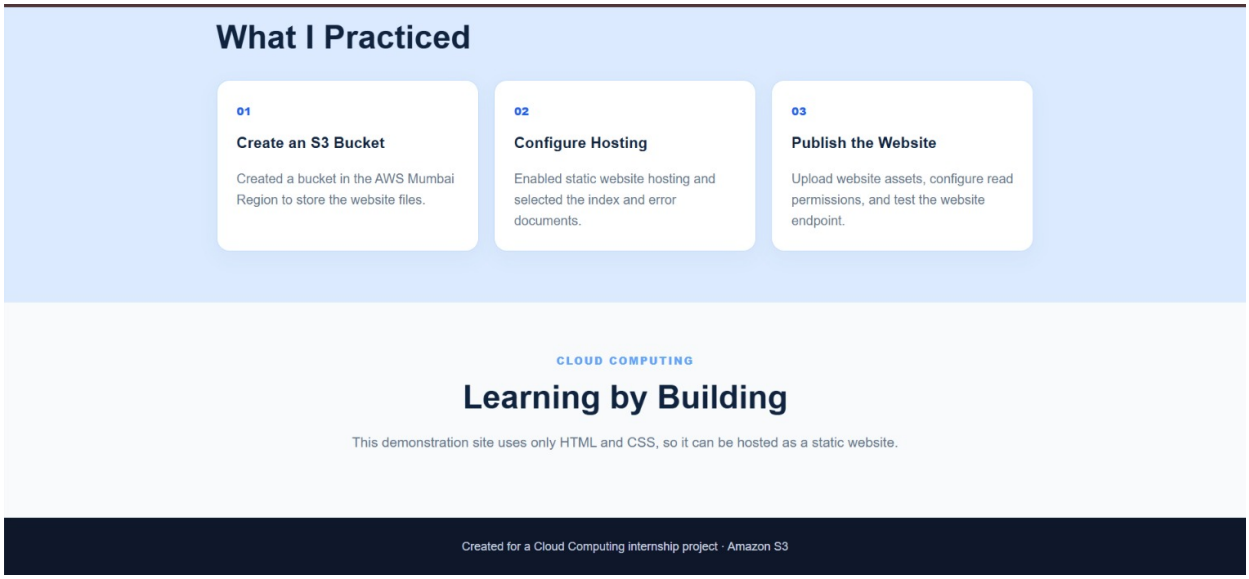


Figure 7: Live website – "Learning by Building" section and footer

5. Result

The static website was successfully hosted on Amazon S3 and verified through the S3 website endpoint. All task requirements were met: bucket created, files uploaded, static hosting enabled with index and error documents, public read access configured, and the live site tested.

Live URL: http://manish-static-site-111026.s3-website.ap-south-1.amazonaws.com

The endpoint works only while the bucket exists and its hosting configuration and permissions remain unchanged.

6. Security Considerations

- The bucket is intentionally public for reading – store only non-sensitive, public content in it.
- The policy grants read-only access (s3:GetObject); no write or delete access is exposed.
- S3 website endpoints support HTTP only. HTTPS requires Amazon CloudFront in front of the bucket.
- Remove public access when the site is no longer needed.

7. Optional Enhancements (not part of this task)

- **Amazon CloudFront:** HTTPS, caching and lower latency.
- **Route 53 custom domain:** a friendly domain name for the site.
- **AWS Amplify Hosting:** the managed option recommended by the console.

8. Key Learnings

- S3 can serve a static site directly through a website endpoint, with no server to manage.
- Index and error documents control the default page and the not-found page.
- A bucket policy, together with Block public access settings, determines whether anonymous visitors can read the files.
- The website endpoint is region-specific: bucket-name.s3-website.region.amazonaws.com.